School of Computer Science and Engineering, UNSW

Week 1 : Course Introduction

Software Architecture for Blockchain Applications, COMP6452

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Never Stand Still

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- Welcome to 2025 Term 2 offering of COMP6452, Software Architecture for Blockchain Applications.
- I hope you are excited to get started.
- Let’s start with a brief course overview

Acknowledgement: Course materials are based on previous offerings of COMP6452, which were originally developed by the Architecture and Analytics Platforms (AAP) team at CSIRO's Data61

https://research.csiro.au/data61/architecture-analytics-platforms/

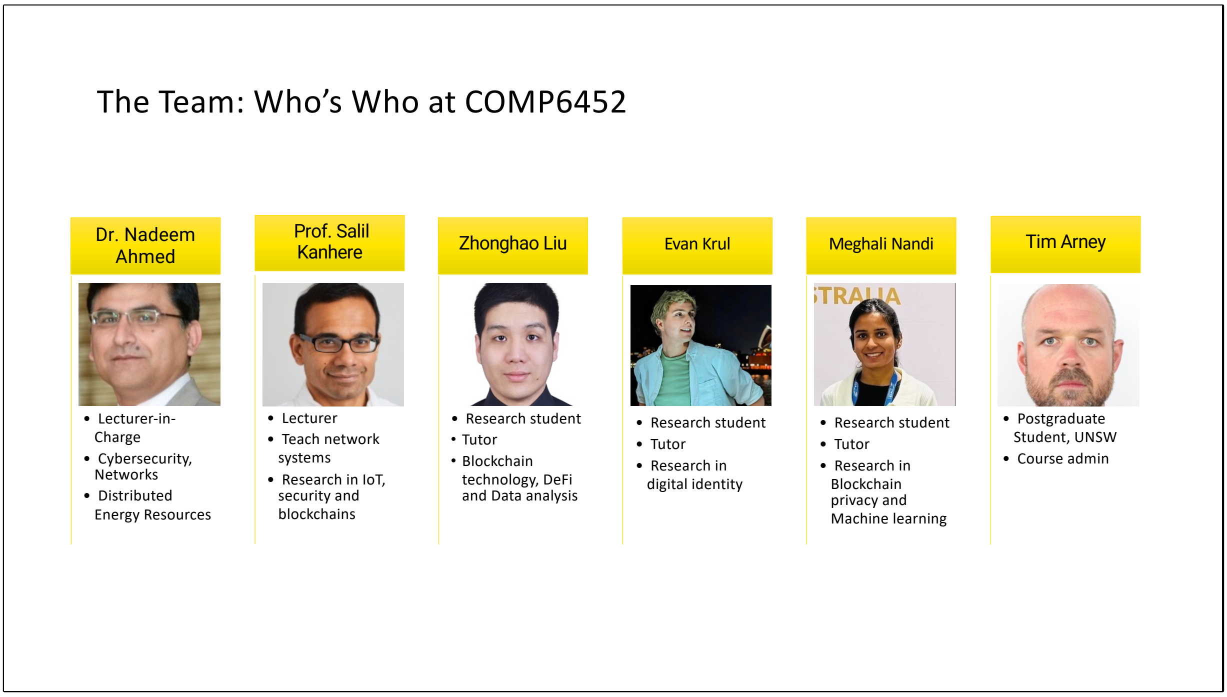
Today's Agenda

Part 1

- Introduction to administrative matters
- What is this course about?

Part 2

- Crypto building blocks



- Ok, the team. What a bunch of good-looking people!
- The main lecturer-in-charge is responsible for the day-to-day running of the course, we have two co-lecturers who will take turn to deliver the weekly topics
- Then we have a highly dedicated team of teaching assistants and course administrator.

Resources

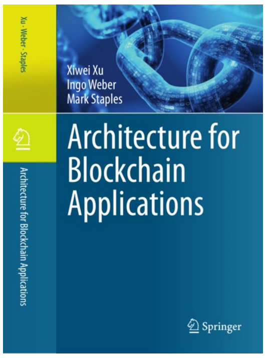
Course website is on WebCMS3

- https://webcms3.cse.unsw.edu.au/COMP6452/25T2/
- Lecture notes
- Link to Echo360 video recordings
- Lab schedules and locations
- Project and Lab exercises
- Link to the Discourse discussion forum

Announcement: Your responsibility to check the announcement/notices on regular basis for important updates/changes to schedule, etc.

Your active participation and interaction is crucial to ensure that all of us get the most out of this course

Textbook – Architecture for Blockchain Applications



Xiwei Xu, Ingo Weber, Mark Staples
Architecture for Blockchain Applications
Springer, 2019

https://link.springer.com/book/10.1007/978-3-030-03035-3

(Access through UNSW Library)

- Supplementary readings
- Some additional notes on slides

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- This book discusses all the important issues of building blockchain applications from software engineering and architecture point of views.
- Sherry (the lead author) was the lecturer-in-charge of this course before and taught this course for 3 years.
- The course content closely follows the textbook, except for more recent topics. The book should be available from the UNSW library.

Course Outline

Week	Date	Monday	Wednesday	Assessment
1	June 2 & 4	• Introduction • Crypto Building Blocks	• Blockchain Basics	Project 1 release Project 2 release
2	June 9 & 11	• Pub Holiday (King's Bday)	• Blockchain Platforms	Project 2 group formation
3	June 16 & 18	• Smart Contracts and Oracles	• Software Architecture Basics	Project 2 Task 1
4	June 23 & 25	• Roles Blockchain Can Play	• Blockchain Taxonomy	Project 1 submission
5	June 30 & 2 July	• Decentralised Application Design Process	• Project 2 – Presentation 1 (use case, requirements, & high-level design)	Project 2 – Task 2 Quiz 1

- Now let's look at the weekly schedule of the course in terms of the topics and assignment activities.
- This time offering has changed a bit from the past offerings of this course. The main change is the topics and their depth of coverage.

- Week 1
  - This week, after the course intro, you will learn the cryptography building blocks followed by basics of blockchains and the various problems you need to understand in applying blockchain technology.
  - Both Projects 1 & 2 are released this week, but they are due on different dates. Note that Project 2 has 3 tasks and 2 assessment components
- Week 2
  - Monday Week 2 is a public holiday.
  - You will learn different blockchain platforms that are currently available. These include Bitcoin, Ethereum and HyperLedger. Group formations for Project 2 is due this week.
- Week 3
  - We start the week with the basics of smart contracts – the programming paradigm that everybody uses to build applications on blockchains

    - This smart contract topic will be the core of your first project. You will be guided through how to write your first smart contract, and how to make it run on a blockchain – then we will give you a few extra activities you need to complete.
    - As mentioned, the course discusses a lot of software architecture-related topics, so starting from Week 2, we will introduce some basics of the software architecture area and related concepts, and then we will talk about software architecture topics in the context of blockchains.
- Week 4
    - We will discuss the role of blockchain in a software architecture. We discuss about the process of developing blockchain-based applications. And how you may decide if blockchain technology is suitable for given situations or not.
    - The taxonomy will allow you to break down the design choice issues into different categories and available options. Having something like this allows you to evaluate and compare different design choices clearly.
- Week 5
    - The first lecture is on decentralised application design process. In the 2nd lecture, we will ask you to present your project 2 design solution.

Course outline – (cont)

Week	Date	Monday	Wednesday	Assessment
6	Flexibility Week			
7	July 14 & 16	• Design Exercise	• Design Patterns for Blockchain Applications	
8	July 21 & 23	• Smart Contract Testing	• Functionality, Performance, and Reliability	
9	July 28 & 30	• Security, Privacy, and Interoperability	• Project 2 – Presentation 2 (design & demo)	Project 2 Task 3 Quiz 2
10	Aug 04 & 06	• Project 2 – Presentation 2 (design & demo)	• Project 2 – Presentation 2 (design & demo)	Project 2 – Task 3 Project 2 submission

- Week 6
  - This is a quiet week – so you can catch up with stuff.
- Week 7
  - We begin this week by following a design exercise. We will discuss design patterns. Design patterns in software engineering is introduced to provide a good solution to recurring or common problems in software development. We will show you some application design patterns that involve blockchains and what sort of problems they tackle. Hopefully, you will be able to identify some of them in your own projects.

- Week 8
  - We will talk about testing smart contracts, again things to consider when testing smart contracts, how to test, etc.
  - We will talk about the performance of blockchain-based systems, general issues and how to, to measure when we talk about performance issues in blockchains.
  - Next, we move onto topics what we call non-functional properties of software, topics like security, reliability, privacy, etc.
  - We will also talk about other aspects like cost of operating and transacting

on a blockchain

- Week 9
  - We discuss security & privacy in the context of blockchains.
  - 2nd lecture is reserved for project 2 Task 3 presentations
- Week 10
  - We will summarise the course and give you some sample questions for the final exam.
  - In the 2nd lecture, we will ask you to present your project 2 demo.

Assessments

Assessment Title	Assessment Type	Marks
Projects	2 projects where you will analyse & implement use cases using blockchains & smart contracts Project 1 individual, Project 2 group (4-5 students)	15 + 30
Quizzes (2)	Quiz Test 2 online quizzes on material covered up to that point	10 x 2 = 20
Final Exam	Central online open-book exam via Inspera A written examination, testing all course content Students must obtain a **passing grade on the exam** to pass the course (i.e., < 50% of the exam points means failing the course)	35

Projects
- Project 1 is an individual assessment where you will develop a smart contract and add some additional functionalities. This project accounts for 15 marks
- 2nd project will be a group activity. The main idea is for you to build a non-trivial blockchain application from scratch, applying the design concepts you have learned in the course. There will be 2 checkpoints for this project, the first one, you need to show us your design and requirement analysis of your application, and then in the final checkpoint, you will be presenting the complete working application to us. This project accounts for 30 marks split as 16 + 14

Quizzes
- There will be 2 online quizzes on Week 5 and 9.
- It consists of a mix of MCQ, True/False, short answer, etc., questions

Final exam
- It will be a 24-hour online open book exam conducted via Inspera
- Once started, you need to complete the final exam within 3 hours
- Note that you must obtain a **passing grade on the exam** to pass the course (i.e., < 50% of the exam points means failing the course)

Getting Help

Discourse discussion forum
- https://discourse01.cse.unsw.edu.au/25T2/COMP6452/

Class consultations
- Lectures – Immediately after class or by appointment
- TAs – During labs or by appointment

Lab classes
- Per timetable advertised in WebCMS
- Help sessions on technical problems & group project work
- Ignore your lab registration, you can attend any lab classes open for the week

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- Use Discourse Forum for basic questions
  - It will help others with similar questions too
  - Also, you may answer each other's questions

- Class consultations
  - Lectures – Immediately after class or by appointment
  - TAs – During tutorials/labs or by appointment

- Lab classes = Lab classes each week (tutors will be present). Ignore your class registration time for labs, you can attend any of the lab classes open each week (and as many times as you'd like). Consider them as your help sessions and "quality" time with the tutors

What is this course about?

Two distinct components

Blockchain

- A general introduction to the topic and to existing blockchain platforms including Bitcoin, Ethereum, and Hyperledger Fabric;
- Smart contracts programming

Software architecture and Blockchain applications

- The functional aspects of software architecture are covered, describing the main roles blockchain can play in an architecture, as well as its potential suitability and design process;
- Non-functional aspects of blockchain applications, which are often cross-cutting concerns including performance, security etc.

Learning Outcomes

On completion of the course, you will be able to:

1. Explain the principles of blockchain & its roles in an application architecture
2. Assess the suitability of blockchains to solve a constrained problem
3. Design blockchain-based software architectures
4. Make functional and non-functional trade-offs for blockchain-based applications
5. Build small applications on a blockchain

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The learning outcomes. These are what we hope you will have learned at the end of the course.

After the course, we want you to be able to:

- we want you to understand the principles of blockchains and explain them clearly, and the roles the technology can play in an application architecture
- You should be able to decide when blockchains are suitable in an application and how to design applications using blockchains
- you should be able to draw up pros and cons of a design in terms of functional and non-functional properties of the system and make a decision based on the trade-offs from the analysis.
- Finally, develop small, but non-trivial applications on blockchains. That is, we would like you to have some hands-on skills after this course.